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PERSPECTIVE

Measuring Food Insecurity

Christopher B. Barrett*

Food security is a growing concern worldwide. More than 1 billion people are estimated to lack sufficient dietary energy availability, and at least twice that number suffer micronutrient deficiencies. Because indicators inform action, much current research focuses on improving food insecurity measurement. Yet estimated prevalence rates and patterns remain tenuous because measuring food security, an elusive concept, remains difficult.

The 2008 global food price crisis, which sparked riots in more than two dozen countries, rekindled political and scientific interest in food security. In their July 2009 joint statement, the G8 heads of state agreed "to act with the scale and urgency needed to achieve sustainable global food security" (1). To direct scarce resources to where they can do the greatest good, actions must be guided by reliable information as to who is food insecure, where, when, and why. This requires improved measurement of food insecurity and its causes and greater attention to key institutional and policy lessons learned.

An Elusive Concept

Among the various definitions currently in use, the prevailing definition, agreed upon at the 1996 World Food Summit, holds that food security represents "a situation that exists when all people, at all times, have physical, social and

economic access to sufficient, safe and nutritious food that meets their dietary needs and food preferences for an active and healthy life." This high standard encompasses more than just current nutritional status, capturing as well vulnerability to future disruptions in access to adequate and appropriate food (2, 3).

Food security is commonly conceptualized as resting on three pillars: availability, access, and utilization. These concepts are inherently hierarchical, with availability necessary but not sufficient to ensure access, which is, in turn, necessary but not sufficient for effective utilization (4). For most of human history, lives were short and unhealthy due in large measure to insufficient macronutrient (carbohydrate, fat, and protein) intake. Beginning in the 18th century, however, a succession of countries broke free of the nutritional poverty trap (5, 6), thanks largely to increased food availability made possible by advances in agricultural production; hence, the common association of food security with supply-side indicators, typically measured in daily calories per person.

Adequate availability is necessary, but does not ensure universal access to "sufficient, safe

and nutritious food." Access is most closely related to social science concepts of individual or household well-being: What is the range of food choices open to the person(s), given their income, prevailing prices, and formal or informal safety net arrangements through which they can access food? As Nobel Laureate Amartya Sen wrote, "starvation is the characteristic of some people not *having* enough food to eat. It is not the characteristic of there *being* not enough food to eat. While the latter can be a cause of the former, it is but one of many *possible* causes" (7). Access reflects the demand side of food security, as manifest in uneven inter- and intrahousehold food distribution and in the sociocultural limits on what foods are consistent with prevailing tastes and values within a community. Access also accentuates problems in responding to adverse shocks such as unemployment spells, price spikes, or the loss of livelihood-producing assets. Through the access lens, food security's close relationship to poverty and to social, economic, and political disenfranchisement comes into clearer focus. But because access is an inherently multidimensional concept, measurement becomes more difficult than with availability (4).

Utilization reflects concerns about whether individuals and households make good use of the food to which they have access. Do they consume nutritionally essential foods they can afford, or do they choose a nutritionally inferior diet? Are the foods safe and properly prepared, under sanitary conditions, so as to deliver their full nutritional value? Is their health such that they absorb and metabolize essential nutrients? Utilization concerns foster greater attention to dietary quality, especially micronutrient deficiencies associated with inadequate intake of essential minerals and vitamins.

Department of Applied Economics and Management, Warren Hall, Cornell University, Ithaca, NY 14853-7801, USA.

*To whom correspondence should be addressed. E-mail: cbb2@cornell.edu

In practice, analysts use proxy measures for different aspects of food security. Choice among indicators necessarily involves trade-offs, so the objective necessitating measurement commonly drives the choice of indicator.

Measures based on higher-cost individual and household surveys can associate measures with targetable individual characteristics, offering depth in measuring two or three of the pillars. Examples include the coping strategies index (8) and food expenditure and dietary diversity (9) measures, which rely on household or individual responses to survey questions about approaches to respond to shocks and past consumption, respectively. Likewise, hunger refers to the physical discomfort caused by a lack of food and can only be properly gathered at the individual level. Underweight summarizes individual anthropometric measures—such as weight-for-height, weight-for-age, or mid upper-arm circumference—at least two standard deviations below global reference values. Undernutrition reflects insufficient dietary energy (caloric) intake, according to internationally agreed standards (10). Malnutrition refers to undernutrition, obesity, and micronutrient (mineral and vitamin) deficiencies.

Simpler food availability measures enable frequent and geographically broad estimates, but at the expense of neglecting waste and the inevitably unequal distribution and uses of food within a population. The low cost and relative ease of generating availability estimates, especially at a national and global scale, account for the lasting appeal of such measures long after the food security community fully absorbed Sen's message.

Measurement matters for at least three major reasons. First, each measure captures and neglects different phenomena intrinsic to the concept of food security, thereby subtly influencing prioritization among food security interventions. Historically, reliance on national food availability estimates has focused attention on food aid shipments and agricultural production strategies to increase food supplies in the short and long term, respectively. Over roughly the past quarter century, Sen's core thesis—that food access accounts for most food insecurity—has focused increased attention on individual-specific hunger and underweight data, which naturally reinforces strategies based on poverty reduction, food price, and social protection policies. For example, the voluntary guidelines on the right to ade-

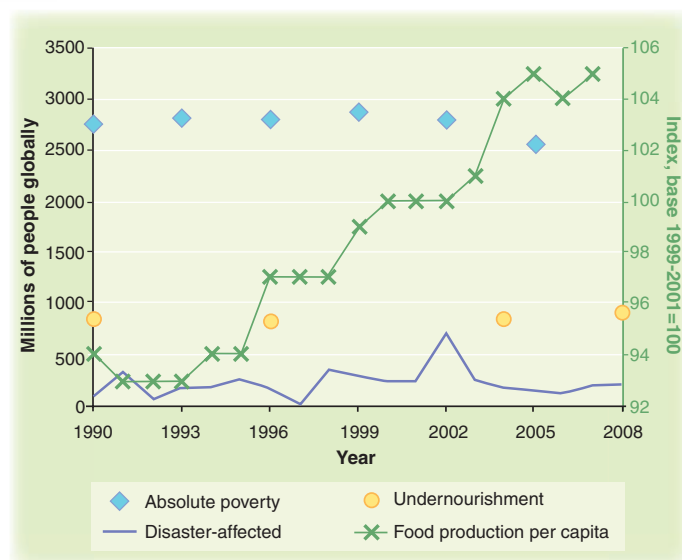


Fig. 1. Different food security proxy indicators paint different pictures. Undernourishment is far greater than just those affected by disasters but much less than those living in absolute poverty. And none of those measures show the improvement over time that food availability measures do. Data sources: Absolute poverty (<\$2/person per day): (31); Undernourishment: (32); Disaster-affected populations: (33). Gross food production per capita (base 1999–2001 = 100) (34). Food production series to be read against right-hand vertical axis; all other series against left-hand vertical axis. Unfortunately, no comparable series are available for micronutrient malnutrition or for perceptions-based measures.

quate food, unanimously agreed by all U.N. Food and Agriculture Organization (FAO) member states in 2004 (11), was a response to lack of progress in individual-level indicators despite growth in aggregate food supplies and incomes.

Second, observational data necessarily report on the past. But policy-makers are most interested in the likely future effects of prospective interventions. An ideal food security indicator would reflect the forward-looking time series of probabilities of satisfying the access criteria (3). Yet there has been little effort to date in testing the forecasting accuracy of currently available indicators (12).

Third, national-level measures inherently lend themselves only to addressing national-scale food availability shortfalls, not intranational access and utilization concerns. Insofar as food insecurity measures diagnostically inform actions, they must be readily associated with targetable characteristics of vulnerable households and individuals and remediable causal factors that lead to food insecurity. The research frontier therefore revolves around the development of cross-nationally comparable, longitudinal monitoring and analysis at the household and individual level.

Patterns and Trends

The most widely cited food insecurity figures are the “undernourishment” estimates generated by

FAO, derived from national-level food balance sheets and strong assumptions about intranational distribution of food. Alternative measures, such as those reported to Congress each year by the U.S. Department of Agriculture (USDA), generated by simulation models based on prices and national accounts and production equations, often differ radically from FAO estimates. For example, in June 2009, FAO estimated that the number of undernourished people globally climbed to 1020 million (13) (Fig. 1); in the same month USDA estimated only 833 million food insecure (14). And the FAO estimate for Asia was more than two-thirds higher than USDA's (642 versus 379 million) while that for Sub-Saharan Africa was nearly one-third lower (265 versus 385 million). Such discrepancies make even macroscale geographic targeting difficult for policy-makers.

The global figures mask considerable heterogeneity among and within regions, especially in trends. In China and Southeast Asia, tens of millions fewer people suffer undernutrition than a generation ago due to broad-based, rapid economic growth. In other regions, including South Asia and parts of east and southern Africa, undernutrition rates have fallen even while the number of undernourished has increased due to population growth. And in some regions (e.g., central Africa), both numbers and rates have increased (15).

Continued reliance on contested national food availability measures reflects the limited availability and timeliness of household and individual data collected in nationally representative surveys. A growing literature proves the value of survey data that capture objective dietary, economic, and health indicators as well as subjective measures of adequacy, risk exposure, and sociocultural acceptability (16–18). Food security measures based on household and individual data routinely generate higher estimates of food insecurity than those derived from more aggregate data. The differences seem attributable to differences not only in intra- and interhousehold nutrient distribution but also in the resulting estimates of nutrient availability (17). Not surprisingly, survey-based estimates of food insecurity are more strongly correlated with poverty estimates, which are likewise generated from household survey data.

Beyond the increased precision that more disaggregated evidence allows, individual- and household-level survey-based measures permit reasonably accurate prediction of who is most

likely to be affected adversely by potentially harmful shocks, such as food price increases, drought, or slumping demand for wage labor. Survey data-based predictions of community-level variables, such as child undernutrition, can even underpin catastrophe insurance contracts that trigger payouts when most needed (19). By contrast, aggregate food availability is a poor predictor of other food insecurity indicators. For example, the undernourished population has increased by 9% globally despite a 12% rise in global food production per capita since 1990 (Fig. 1).

Although the most severe food insecurity is typically associated with disasters such as drought, floods, war, or earthquakes, most food insecurity is associated not with catastrophes, but rather with chronic poverty (Fig. 1). Only 8% of hunger-related deaths worldwide in 2004 were caused by humanitarian emergencies; 92% were associated with chronic or recurring hunger and malnutrition (20). Similarly, in every country, rates of child stunting—reflecting chronic undernutrition—far exceed those of child wasting—indicating short-term, acute undernutrition—with the difference greatest in the poorest countries.

Because most food insecurity is seasonal or regular but aperiodic—i.e., associated with temporary unemployment, episodes of ill health, or other recurring adverse events—people anticipate such possibilities and routinely engage in precautionary behaviors to try to mitigate their risk. Hence perceptions-based survey measures consistently find food insecurity rates several times higher than related hunger or insufficient-intake measures (21).

Even perceptions data may not suffice to capture utilization problems, such as those associated with micronutrient malnutrition. The prevalence of micronutrient deficiencies is imprecisely known; rough estimates suggest that iodine, iron, vitamin A, and zinc shortfalls alone affect at least 2 billion people, disproportionately women and children. This leads to increased risk of both chronic and infectious disease, aggravates diseases' effects, and leads to irreversible loss of cognitive and physical function, especially during the crucial period from –9 to 24 months of age, during which children are biologically vulnerable and completely dependent on caregiver knowledge to utilize foods appropriately (22–25). These irreversible effects foster persistent poverty, reinforcing the consequences of food insecurity.

Policies and Institutions for Effective Intervention

Effective, direct food security interventions depend on effective targeting of the vulnerable subpopulation(s) and of the causes of insecurity, as well as prompt response. Where data collection is timely, causal factors can be robustly associated with food insecurity measures,

and when predictive models have demonstrable accuracy, preventive measures can substantially reduce unnecessary human suffering. The long-term consequences of crises can be limited where appropriate policies and institutions are already in place, such as social protection schemes to cushion people in times of adversity and early childhood health programs to protect the most vulnerable from even short-lived interruptions in essential nutrient intake.

Automatic stabilizer and safety net programs are important means of circumventing inconsistent or inadequate government and donor response. Political and economic elites are rarely severely affected by crises, nor do they suffer chronic food insecurity. Because they rarely face intense, immediate political pressure, slow, halting or incompetent government and donor response is a recurring problem. For example, the median delivery time for emergency food aid from the United States, the main global food donor, is nearly 5 months due to legislative restrictions on procurement and shipping (26). And even prescient early-warning systems often go unheeded. In the Niger crisis of 2004–2005, for example, below-normal rains and anticipated locust attacks led to a low cereals harvest that elicited prompt government and United Nations appeals for emergency assistance in November 2004. But the global response was anemic. By July 2005, the Niger situation was finally attracting graphic global media coverage that led to a significant global response, much of which arrived with the next harvest.

These delays are both deadly and expensive. In Niger, quite apart from the still unclear human health toll and lives lost to delays, the cost per beneficiary for World Food Programme deliveries more than tripled from February to August 2005, from \$7 to \$23, due to far greater need for supplemental and therapeutic foods instead of cheaper, bulk commodities, and the need for airlift and other quicker, but more expensive, logistical support (19). Poorly conceptualized or implemented relief programs can adversely affect communities, leaving them more vulnerable to food insecurity by displacing commercial food trade, affecting local prices, or distorting incentives and behaviors; hence, the need for prearranged financing facilities, social protection programs such as employment guarantee schemes, and other ready-made responses to emergent food security crises.

Perhaps the most important factor determining the efficacy of food security interventions is the quality of targeting. Does assistance reach the intended beneficiaries? Good targeting is exceedingly difficult. The neediest individuals are not always easily identified, even in participatory or community-based targeting efforts, for example, due to social isolation or discrimination. Effective targeting is usually based on a mixture of geographic indicators, observable

individual or household characteristics, program restrictions (such as work requirements) that induce self-selection out of participation by the non-needy, or community consultations. Identifying the most needy and the optimal form of assistance involves trade-offs across time, efficacy, and cost and commonly requires triangulation using multiple indicators across time and levels of aggregation (3, 26). In responding to transitory food insecurity associated with sudden natural disasters (e.g., earthquakes, hurricanes), careful and expensive data collection may be inappropriate, especially for short-lived interventions (27). By contrast, careful targeting is essential for long-term programs that address more common chronic food insecurity.

The greatest food security gains typically come not directly, from feeding programs, but rather indirectly, through policies that promote poverty reduction through employment creation and productivity growth among the poor, as well as safety nets to safeguard the vulnerable nonpoor. Enhanced control over productive assets and access to the technologies and markets necessary to sustainably use them to generate a stable livelihood are especially crucial to reducing vulnerability to food insecurity and facilitating the escape from poverty traps (28); hence, the importance of continued efforts to boost crop productivity, especially for micronutrient-rich foods, where food availability remains limiting, as is true of dozens of low-income countries.

Conclusions

Measurement drives diagnosis and response. As global attention returns to food security, new opportunities emerge to improve its measurement. Research is appropriately and increasingly moving toward survey-based anthropometric and perceptions measures to improve the disaggregated identification of food-insecure subpopulations and their targetable characteristics and behaviors. But the greatest advances in the measurement of food insecurity will come from three developments. First, a global network of sentinel sites using a standardized core survey protocol for regular, repeated household- and individual-level monitoring would enable us to track the coevolution of multiple food security indicators with targetable individual, household, and community characteristics across continents and to rigorously monitor and evaluate the impacts of various policy and project interventions. The recent review of social sciences within the CGIAR calls for this, modeled in part on the National Science Foundation's Long-Term Ecological Research network (29). Second, if we knew better the predictive accuracy of different indicators in forecasting future food security states, we could more cost-effectively concentrate data collection on measures of which targetable actions can be most reliably programmed.

This would help overcome the breadth-versus-depth trade-offs that presently limit the availability of suitable time series data at the individual and household level. Finally, just as poverty research is moving beyond static, snapshot measures to dynamic mobility ones, especially with respect to critical behavioral thresholds (28), so must the food security research community begin developing measures based on longitudinal data that capture the risk of food insecurity that respondents routinely voice in perceptions-based measures.

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PERSPECTIVE

Precision Agriculture and Food Security

Robin Gebbers^{1*} and Viacheslav I. Adamchuk²

Precision agriculture comprises a set of technologies that combines sensors, information systems, enhanced machinery, and informed management to optimize production by accounting for variability and uncertainties within agricultural systems. Adapting production inputs site-specifically within a field and individually for each animal allows better use of resources to maintain the quality of the environment while improving the sustainability of the food supply. Precision agriculture provides a means to monitor the food production chain and manage both the quantity and quality of agricultural produce.

To secure food supplies for the future requires adequate quantities and quality of agricultural produce, intensive yet envi-

ronmentally safe production, and the sustainability of the resources involved. In addition, the ability to track food materials from production through processing, storage, and retail provides added capability to respond to changing market conditions, ensure proper food nutrition and safety, and affect national and international policies related to food security.

Precision agriculture, or information-based management of agricultural production systems,

emerged in the mid-1980s as a way to apply the right treatment in the right place at the right time (1–3). Increasing awareness of variation in soil and crop conditions, combined with the advent of technologies such as global navigation satellite systems (GNSSs), geographic information systems (GISs), and microcomputers, serve as the main drivers (1, 2). Initially, precision agriculture was used to adapt fertilizer distribution to varying soil conditions across an agricultural field. Since then, additional practices have evolved, such as automatic guidance of agricultural vehicles and implements, autonomous machinery and processes, product traceability, on-farm research, and software for the overall management of agricultural production systems.

Apart from field crop production, precision agriculture technologies have been applied successfully in viticulture and horticulture, including orchards, and in livestock production, as well as pasture and turf management. Applications range from the tea industry in Tanzania and Sri Lanka to the production of sugar cane in Brazil; rice in China, India, and Japan; and cereals and sugar beets in Argentina, Australia, Europe, and the United States (4). Despite differences in the types of technology and the areas of adoption, the goals of precision agriculture are threefold. First, to optimize the use of available resources to increase the profitability and

¹Department of Agricultural Engineering, Leibniz-Institute for Agricultural Engineering (ATB), Max-Eyth-Allee 100, D-14469 Potsdam, Germany. ²Biological Systems Engineering Department, University of Nebraska-Lincoln, 203 Chase Hall, Lincoln, NE 68583-0726, USA.

*To whom correspondence should be addressed. E-mail: rgebbers@atb-potsdam.de